Lefschetz Hyperplane Section Thms:

Lem. $i: D \hookrightarrow X$ hyperplane section (X: projective var./$\mathbb{C}$). Then $\forall F^\bullet \in Perv(X)$,

i) restriction map $\mathbb{H}^k(X, F) \longrightarrow \mathbb{H}^k(D, i^*F)$ is an isom. for $k<-1$ & injective for $k=-1$.

ii) pushforward map $\mathbb{H}^k(D, i^!F^\bullet) \longrightarrow \mathbb{H}^k(X, F)$ is an isom. for $k>1$ & surj. for $k=1$.

Pf:

$$U \overset{j}{\hookrightarrow} X \overset{i}{\hookleftarrow} D$$

$U = X - D$

$$j_!j^!F \longrightarrow F \longrightarrow i_*i^*F \xrightarrow{[1]} \rightsquigarrow \cdots \longrightarrow \mathbb{H}^k_c(U; j^*F) \longrightarrow \mathbb{H}^k(X, F) \longrightarrow \mathbb{H}^k(D, i^*F)$$

$j^*F \in Perv(U) \overset{\text{Artin vanishing}}{\Longrightarrow} \mathbb{H}^k_c(U, j^*F) = 0$ for $k<0$.

$\overset{\text{les}}{\Longrightarrow}$ (i)

Use dual sequence $i_!i^!F \longrightarrow F \longrightarrow Rj_*j^*F \xrightarrow{[1]}$ to get (ii)

Thm (Lefschetz hyperplane section for IC). $X \subset \mathbb{P}^N$: projective, cx variety

H: generic hyperplane & $D = X \cap H$. Then

$$IH^i(X, \mathbb{Q}) \longrightarrow IH^i(X \cap H, \mathbb{Q}) \quad \text{is isom.}$$

for $0 \le i \le n-2$ & monomorphism for $i = n-1$.

Pf: $i: D \hookrightarrow X$. Previous Lemma implies

$$IH^{k+n}(X, \mathbb{Q}) := \mathbb{H}^k(X; IC_X) \longrightarrow \mathbb{H}^k(D; i^*IC_X) \text{ is}$$

isom $\forall k<-1$ & monom. for $k=-1$.

Claim: $i^*IC_X \cong IC_D[1]$

Why: D is "normally singular", i.e.

Therefore: $\mathbb{H}^k(D, i^*IC_X) = \mathbb{H}^k(D, IC_D[1]) = IH^{k+n}(D, \mathbb{Q})$ //

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First show HL implies splitting of ord. shvs

Hard Lefschetz for IC: (lead w/ Deligne's Thm for smooth decomp^n thm.)

$f: X \to Y$ projective. $\mathcal{L}$: line bundle on X s.t.

"$\mathcal{L}$ is f-ample": $\mathcal{L}|_{f^{-1}(y)}$ is ample $\forall y \in Y$.

e.g. $X \xhookrightarrow{i} \mathbb{P}^n \times Y \to Y$; $\mathcal{L} := i^*(\mathcal{O}(1) \boxtimes \mathcal{O}_Y)$ works.

$L = c_1(\mathcal{L}) \in H^2(X; \mathbb{Q})$ first Chern class

[If $0 \to \mathbb{Z} \to \mathcal{O}_X \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}_X^* \to 1$ is exp^n ses, then

$$\to H^1(X, \mathcal{O}_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z})$$
$$\psi \qquad\qquad \psi$$
$$\mathcal{L} \longmapsto L$$

: connecting map]

L determines $L: \underline{\mathbb{Q}}_X \to \underline{\mathbb{Q}}_X[2]$

bc $\mathrm{Hom}_{D^b(X)}(\underline{\mathbb{Q}}_X, \underline{\mathbb{Q}}_X[2]) = \mathbb{H}^2(X, \underline{\mathbb{Q}}_X) = H^2(X, \mathbb{Q})$

$\otimes IC_X$ ↝ $IC_X \to IC_X[2]$

$\Rightarrow$ $Rf_* IC_X \to Rf_* IC_X[2]$

$\Rightarrow$ $L: {}^p\mathcal{H}^i(Rf_* IC_X) \to {}^p\mathcal{H}^{i+2}(Rf_* IC_X)$] when X sm. & $f: X \to pt$, this is cup pr

$\Rightarrow$ $L^i: {}^p\mathcal{H}^{-i}(Rf_* IC_X) \to {}^p\mathcal{H}^i(Rf_* IC_X)$ (★)

Thm. (Relative Hard Lefschetz). $f: X \to Y$: projective, L as above, then (★) is isom. $\forall i$.

Ex: $f: X \to Y$ proj. btwn sm. varieties of rel. dim d.

$R^{d-i} f_* \underline{\mathbb{Q}}_X \cong R^{d+i} f_* \underline{\mathbb{Q}}_X$.

Ex: X: projective, $f: X \to pt$.

$L^i: IH^{n-i}(X, \mathbb{Q}) \xrightarrow{\sim} IH^{n+i}(X, \mathbb{Q})$ $\forall i > 0$

Proofs of Hard Lefschetz:

- reduction to pos. char. via ℓ-adic shvs (BBD)
- polarizable Hodge modules (Saito)
- de Cataldo-Migliorini (classical Hodge theory & hyperplane sections, etc.)

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Hard Lefschetz as an $\mathfrak{sl}_2$ action:

$V = \bigoplus_{k=0}^{2n} V^k$, $L: V \to V^{k+2}$ (& $H = \bigoplus_k H|_{V^k}$)

Pat $H|_{V^k} = (k-n)\cdot 1_{V^k}$. Then TFAE:

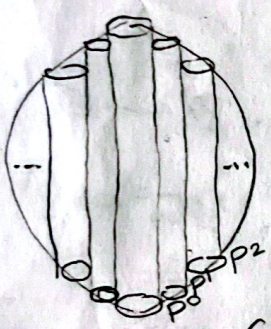
① $L^{n-k}: V^k \xrightarrow{\sim} V^{2n-k}$ is iso. $\forall k$

② $\exists\ \Lambda: V^k \to V^{k-2}$ $\forall k$ s.t.

$[H,L] = 2L$, $[H,\Lambda] = -2\Lambda$, $[L,\Lambda] = H$.

Pf: $P^k = \ker(L^{n-k+1}: V^k \to V^{2n-k+2}) \subset V^k$ "primitive subsp".

$\Rightarrow V = \bigoplus_{k=0}^{n} \bigoplus_{j=0}^{n-k} L^j(P^k)$



$v \in P^k \rightsquigarrow v, Lv, L^2v, \dots, L^m v$ for $m = n-k$

$\Lambda(L^j v) = j(m-j+1)L^{j-1}\cdot v$

↑ defines Λ everywhere

Can check relations: e.g. $[L,\Lambda](L^j v) = [j(m-j+1) - (j+1)(m-j)]\cdot L^j v$

$= (2j-m)L^j v = H(L^j v)$

Other way: $\mathfrak{sl}_2$-rep $\Longleftrightarrow$ $\oplus$ of irreducible strings

$\Longleftrightarrow L^r: V_{-r} \to V_r$ always isom.

weight$(V^k) = k-n$ so this finishes ① //

Proof of "non-relative" Hard Lefschetz:

Idea: Build $\mathfrak{sl}_2$-rep. completing L. Lowering operator will be dual to L (need a form; use symplectic geometry!)

Assume for simplicity $\mathcal{L}$ is very ample, i.e. $\exists\ X \xhookrightarrow{i} \mathbb{P}^N$ s.t. $\mathcal{L} \cong i^*\mathcal{O}(1)$.

$\mathbb{P}^N$ has a Kähler form ω_{FS} ("Fubini-Study") $([x_0:\dots:x_N]) = \frac{i}{2\pi}\partial\bar\partial \cdot \log\left(\sum_{j=0}^{N} |x_j|^2\right)$

$L = \omega \wedge (-)$ in de Rham coho. Define Λ s.t. $\langle \omega\wedge\alpha, \beta\rangle_{L^2} = \langle \alpha, \Lambda\beta\rangle_{L^2}$.

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~~Proof of Hard Lefschetz for non-relative version:~~

Relative Hard Lefschetz for smooth maps follows from base change. In general: [De Cataldo – Migliorini]

▷ find hyperplane $i: D \hookrightarrow X$ s.t. $\mathcal{O}(D) = \mathcal{L}$.

▷ apply Lefschetz hyperplane thm to reduce to D:

$$P^a = {}^p\mathcal{H}^a(Rf_* \underline{\mathbb{Q}}_X[n])$$
$$Q^a = {}^p\mathcal{H}^a(Rf_* \underline{\mathbb{Q}}_D[n-1])$$

$\rightsquigarrow$ $P^a \xrightarrow{\text{res}} Q^{a+1}$, $Q^a \xrightarrow{\text{Gysin}} P^{a+1}$,

$$P^a \longrightarrow Q^{a+1} \longrightarrow P^{a+2}$$
$$L = i_* i^*(-) = [D] \cup -$$

Relative Weak Lefschetz: $P^a \xrightarrow[(a<-1)]{\sim} Q^{a+1}$, $P^{-1} \hookrightarrow Q^0$

$Q^{a-1} \xrightarrow[(a>1)]{\sim} P^a$, $Q^0 \twoheadrightarrow P^1$

Then $\forall r \geq 2$,

$$P^{-r} \xrightarrow{\sim} Q^{-r+1} \xrightarrow[\text{inductive hypothesis}]{\sim} Q^{r-1} \xrightarrow{\sim} P^r$$

Only remaining is:

$$P^{-1} \xhookrightarrow{\rho} Q^0 \xrightarrow{\gamma} P^1$$

Claim: $Q^0 \cong \text{Im}(\rho) \oplus \ker(\gamma)$

This is the hard part!!

$$\mathcal{X} = \{(x, H) \in X \times (\mathbb{P}^N)^\vee \mid x \in H\} \xrightarrow{\tilde{f}} Y \times (\mathbb{P}^N)^\vee \xrightarrow{p} Y$$

$B = {}^p\mathcal{H}^0(Rf_* \underline{\mathbb{Q}}_{\mathcal{X}}[n+N-1])$, $A = p^* P^{-1}[N]$, $C = p^* P^1[N]$

$\Rightarrow$ $A \xhookrightarrow{\rho} B \xrightarrow{\gamma} C$

Univ. Hyperplane Lemma: Every sub-obj. of B coming from Y (of form $p^* F[N]$) is contained in $\rho(A)$. Every quot. of B coming from Y factors through $B \to C$.

Return to this later: hard enough that it is proved along with decomp^n thm. in parallel!

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